



ELK406E

Distribution of Electrical Energy

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Office Hours: Tuesday 11:30-12:00

Course Description:

- Description of network types according to voltage levels and their configurations.
- Selection criterias of cross-section of lines.
- Calculation of voltage drop in the varius type medium and low voltage distribution systems.
- Economical cross-section calculations of varius type medium voltage distribution systems.



Course Description:

- Power factor correction, benefits gained with compensation and installation of compensation systems.
- Network faults and short circuit current calculations.
- Laboratory work at the electrical Installation laboratory experiments with various protection relays, voltage regulators, Instrument transformers



Course Objectives:

- Learning logical steps of design of distribution systems
- Network Calculations
- Learning principles operation of distribution systems
- Learning principles of reactive power compensation
- Network faults and short circuit current calculations



GRADING SCHEME

Homework/project	10 %
Midterm Exam (2)	50%
Final Exam	40 %

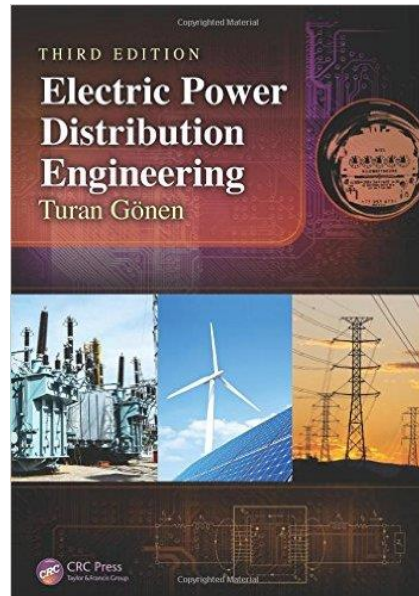
- ✓ *Final exam is a comprehensive exam that will cover the entire semester's material*



Attendance Policy

- Class attendance (70%) is mandatory.
- This course requires an active participation from student.
- You are responsible for all material covered in class, even when you are not present.
- Attendance for the final exam is mandatory.





T. Gonen Electric Power Distribution System Engineering, CRC Press, 2014

- **E.Lakervi. J. Holmes, Electricity Distribution Network Design, IEE Power Eng. Series, Peter Peregrinus,1995**
- **W. H. Kersting, Distribution System Modelling and Analysis, CRC Press, 2002**

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Lecture schedule:

Week	Date	Topics
1	<i>February 18, 2025</i>	Introduction to distribution systems, regulations, standards
2	<i>February 25, 2025</i>	Component of Distribution networks, Underground and Overhead lines, selection criteria of cross-section of conductors
3	<i>March 04, 2025</i>	Description of network types according to voltage levels and their configurations
4	<i>March 11, 2025</i>	Calculations of voltage drop on different configurations of secondary distribution systems
5	<i>March 18, 2025</i>	Calculations of voltage drop on different configurations of secondary distribution systems
6	<i>March 25, 2025</i>	Calculations of voltage drop on different configurations of primary distribution systems
	<i>April 01, 2025</i>	Spring Break
7	<i>April 08, 2025</i>	Calculations of voltage drop on different configurations of primary distribution systems
8	<i>April 15, 2025</i>	Calculations of voltage drop on different configurations of primary distribution systems (Midterm Exam 1)
9	<i>April 22, 2025</i>	Line losses, short circuit calculations
10	<i>April 29, 2025</i>	Short circuit calculations and selection of circuit breakers
11	<i>May 06, 2025</i>	Power factor correction, Reactive Energy Tariff
12	<i>May 13, 2025</i>	Protection system, Instrument Transformers
13	<i>May 20, 2025</i>	Over current Differential protection and Relay coordination (Midterm Exam 2)
14	<i>May 27, 2025</i>	Harmonics and resonant circuits, Smart grids and metering systems, electricity markets

